Manual Chapter : Preventing TCP Connection Requests From Being Dropped

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Preventing TCP Connection Requests From Being Dropped

Overview: TCP request queuing

TCP request queuing provides the ability to queue connection requests that exceed the capacity of connections for a pool, pool member, or node, as determined by the connection limit. Consequently, instead of dropping connection requests that exceed the capacity of a pool, pool member, or node, TCP request queuing makes it possible for those connection requests to reside within a queue in accordance with defined conditions until capacity becomes available.

When using session persistence, a request becomes queued when the pool member connection limit is reached.

Without session persistence, when all pool members have a specified connection limit, a request becomes queued when the total number of connection limits for all pool members is reached.

Conditions for queuing connection requests include:

- The maximum number of connection requests within the queue, which equates to the maximum number of connections within the pool, pool member, or node. Specifically, the maximum number of connection requests within the queue cannot exceed the cumulative total number of connections for each pool member or node. Any connection requests that exceed the capacity of the request queue are dropped.
- The availability of server connections for reuse. When a server connection becomes available for reuse, the next available connection request in the queue becomes dequeued, thus allowing additional connection requests to be queued.
- The expiration rate of connection requests within the queue. As queue entries expire, they are removed from the queue, thus allowing additional connection requests to be queued.

Connection requests within the queue become dequeued when:

- The connection limit of the pool is increased.
- A pool member's slow ramp time limit permits a new connection to be made.
- The number of concurrent connections to the virtual server falls to less than the connection limit.
- The connection request within the queue expires.

Preventing TCP connection requests from being dropped

When you enable TCP request queuing, connection requests become queued when they exceed the total number of available server connections.

1. On the Main tab, click **Local Traffic › Pools** .
   The Pool List screen opens.
2. Click a pool name in the Pool List.
3. From the **Configuration** list, select **Advanced**.
4. In the **Enable Request Queuing** list, select **Yes**.

5. In the **Request Queue Depth** field, type the maximum number of connections allowed in the queue.

   If you type zero (0) or leave the field blank, the maximum number of queued connections is unlimited, constrained only by available memory.

6. In the **Request Queue Timeout** field, type the maximum number of milliseconds that a connection can remain queued.

   If you type zero (0) or leave the field blank, the maximum number of milliseconds is unlimited.

7. Click **Update**.

Connection requests become queued when they exceed the total number of available server connections.

Enabling TCP enhanced loss recovery

Although this feature is disabled by default, you can enable Enhanced Loss Recovery using TCP profiles. This will increase the TCP performance for networks with low RTT and high packet loss. Enhanced loss recovery retransmits lost segments multiple times during a loss recovery period to prevent a timeout recovery.

This feature can be used only when the Selective ACKs feature is enabled for the TCP connection.

1. On the Main tab, click **Local Traffic › Profiles › Protocol › TCP** .
2. Click an existing profile or click **Create**.
3. Scroll down to the Loss Detection and Recovery area, and select the **Custom** check box at the right.
4. For the **Enhanced Loss Recovery** setting, select the check box.

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